# PRAKHAR TRIPATHI

## SOFTWARE ENGINEER | BACKEND & SYSTEMS

*Designing scalable backend systems, open-source frameworks, developer tooling, and distributed platforms with a focus on maintainability and performance.*

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## EDUCATION

**GLA University, Mathura.**
**Bachelor of Technology in Computer Science (Core)**
*August 2022 to May 2026*

## SKILLS

### Fundamentals & Programming
Data Structures & Algorithms (DSA), Object Oriented Programming (OOPs) TypeScript, JavaScript, Java, Python

### Web Technologies
Node.js, React, Angular, Express.js, Spring Boot, Spring MVC, Spring Security, RESTful APIs, Microservices, JWT, OAuth2, RBAC, HTML, CSS (SASS), Responsive UI/UX

### Database Management
MySQL, PostgreSQL, MongoDB (Atlas), Redis (Basics)

### Tools & Frameworks
Git, GitHub, Z Shell, Maven, JUnit, OpenAPI (Swagger), Postman, Razorpay, FFmpeg, RabbitMQ, Raspberry Pi, VS Code, IntelliJ - IDEA

### Cloud & DevOps
AWS (EC2, S3, IAM), Google Cloud (Compute Engine, Cloud Run, IAM), Docker, GitHub Actions, CI/CD, Deployment Pipelines

### Certifications
Google AI Professional Certificate
Google Cloud Infrastructure: Core Services (View Credentials)

### Currently Exploring
Deno (Hono), and Rust (Systems)

## PROJECTS

### SculptorTS - Open-Source Backend Framework & CLI
April 2026 – Present | NPM | GitHub | Full Details | Docs | Releases
TypeScript • Node.js • Express.js • Open Source • Framework Design

- Designing and maintaining SculptorTS, an open-source TypeScript-first backend framework built on Express, focused on package-first application architecture and developer productivity.
- Engineered a modular ecosystem spanning **7 npm packages** with **30k+ combined downloads**, covering runtime, routing, dependency injection, configuration, CLI, code generation, and project scaffolding.
- Designed core framework features including hybrid routing, decorator-driven dependency injection using Reflect Metadata, automated scaffolding, diagnostics, AI-assisted tooling, and native Express builder integration.
- Built a complete CLI ecosystem with project generation, package management, upgrades, diagnostics, and developer tooling to simplify application lifecycle management.

### Pharmetrix - Smart Pharma Inventory & POS (Cold-Chain IoT)
Sept 2025 – Present | Live | **Patent Pending**
Express.js • React • Node.js • MongoDB • Raspberry Pi • Docker

- Developing an IoT-enabled pharmacy inventory and POS platform with batch tracking, expiry-first dispensing, live inventory, and role-based access control.
- Architected a distributed system with 3 backend services (Pi, Gateway, Global) and 2 React frontends, enabling edge processing with centralized cloud management.
- Integrated Raspberry Pi hardware for barcode scanning, camera capture, and temperature monitoring with queued cloud synchronization over Tailscale.

### SpendSmart - Expense Tracker with Data Visualization
April 2026 | GitHub
Spring Boot • React • PostgreSQL • Redis • RabbitMQ • Docker • AWS

- Built a personal finance platform using an 11-microservice distributed architecture supporting authentication, budgeting, transactions, analytics, reporting, and notifications.
- Implemented JWT, OAuth2, RBAC, Redis session caching, and RabbitMQ-driven email, notification, and scheduled reminder workflows.
- Containerized backend services with Docker and deployed them on AWS EC2, with a cloud-portable React frontend.